

NMAO Build — Plain-Language Glossary

A running reference of the technical terms we use as we build, each explained simply and tied to this project. We add to it as new terms come up.

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The tools we work in

CLI (Command Line Interface) — Controlling a program by typing commands instead of clicking.

Example: `supabase functions deploy round-controller`.

Terminal / shell / bash — The text window where you type CLI commands. “Bash” is the most common shell (the language the terminal understands). Same idea as Finder/Explorer, but keyboard-driven.

Supabase — The service hosting our backend: the database, user login, file storage, and edge functions, all in one. Our project lives there.

Postgres (PostgreSQL) — The actual database engine inside Supabase. It stores every table (competitors, entries, pods, results...) and answers questions written in SQL.

Dashboard — The point-and-click website for Supabase, as opposed to the CLI. The SQL Editor and Edge Functions pages live there.

Talking to the database

SQL (Structured Query Language) — The language for asking a database to store or fetch data.

Example: `select count(*) from entries;` means “how many entries are there?”

Schema — The shape of the database: what tables exist and what columns each has. Ours is defined in the migrations folder.

Migration — One file of SQL that builds or changes the schema, numbered so they run in order.

Example: `20260805120000_tournament_engine.sql` creates the rounds/entries/pods tables.

Seed / seed data — Sample data loaded into an empty database so we have something to work with.

`seed_demo.sql` creates a fake tournament round.

Seed output — The text the seed script prints when it runs (e.g. the new `round_id`), so you know it worked and can grab values you need.

Row / record & column / field — A row is one entry in a table (one competitor, one pod). A column is one piece of info on that row (a competitor's name, rank).

UUID — A long, random, guaranteed-unique ID like `a3f8c1e2-9b0d-4f...`. We use them so two rounds (or competitors) can never share an ID.

The engine pieces

Edge function — A small program running on Supabase's servers that does work when called. Ours is `round-controller`, which runs the tournament pipeline.

Deploy — To publish your latest code so the live server runs it. If you change the function's code, you redeploy so the change takes effect.

Invoke / call — To run a deployed function. `supabase functions invoke round-controller ...`

Endpoint / URL — The web address a function answers at. Calling it is sending a request; what it sends back is the response.

POST — The type of web request that sends data to a server (as opposed to GET, which just asks for data). We POST each pipeline step.

Request body — The data attached to a request. Ours is JSON telling the function which round and step to run.

Pipeline / step — The tournament runs in ordered stages (steps): `divide → assign_judges → resolve → distribute`. Each is one job.

Idempotent — Safe to run more than once: running a finished step again does nothing instead of doubling up. Protects us from accidental double-clicks/retries.

Data formats & code

JSON (JavaScript Object Notation) — A simple text format for structured data, written as “key”: value pairs in curly braces. Example: `{"roundId":"a3f8...", "step":"divide"}`.

Variable — A named placeholder for a value. In the terminal, `RID=a3f8...` lets you write `$RID` instead of the long ID afterward.

Flag / option — An extra setting on a command, usually starting with two dashes. Example: `--body "..."` attaches data to the request.

TypeScript / TS — The programming language the engine is written in (a stricter version of JavaScript). Files end in `.ts`.

Key / token / auth — A secret string that proves who's allowed to do something. Two of ours: the **anon key** (public, safe to share, limited by security rules) and the **service_role key** (secret, all-powerful, bypasses the rules — it only ever lives in the server's settings, never in chat or the app).

Version control (Git / GitHub)

Git — A system that tracks every change to the code so you can see history and undo mistakes.

Repo (repository) — The project folder Git is tracking (NMAO-Tournament).

Commit — A saved snapshot of changes with a short note describing them.

Push — Upload your commits to GitHub so they're backed up and shareable.

GitHub — The website that hosts the repo online.